

Report
Of
C++ PROGRAMMING INTERNSHIP

Submitted to
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
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Roll No.: 25SCS1003000683
Batch: 2025-2029

2026-27

CANDIDATE'S DECLARATION

I, ABHISAR SHARMA, hereby declare that this internship report titled “C++ Programming Internship at CodSoft” has been prepared by me as part of the academic internship requirement for the Bachelor of Technology in CSE.

The report presents the work completed during my virtual C++ Programming internship, including the Number Guessing Game, Simple Calculator and To-Do List projects.

I affirm that the work described in this report is based on my internship activities and has not been submitted elsewhere for an academic award.

Signature: _____

ABHISAR SHARMA | 25SCS1003000683

ACKNOWLEDGEMENT

I sincerely thank CodSoft for providing me with the opportunity to complete a C++ Programming virtual internship and work on practical programming tasks.

I am grateful to my faculty members and the School of Computer Science and Engineering, IILM University, for providing the academic guidance and framework required to complete this internship.

I also thank my parents for their continued support and encouragement throughout the internship.

Signature: _____

ABHISAR SHARMA

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4. PROJECT DESCRIPTION

4.1 Introduction

The internship was completed as a C++ Programming virtual internship with CodSoft. It focused on applying programming fundamentals through practical console-based applications.

Three tasks were completed: Number Guessing Game, Simple Calculator and To-Do List. The programs were written, compiled, tested with user input and submitted through GitHub.

4.2 Organization Profile

CodSoft provides project-based learning and virtual internship opportunities designed to give learners practical programming experience. The supplied offer letter identifies the position as a C++ Programming virtual internship and describes it as an educational opportunity focused on learning, skill development and hands-on knowledge. filecite turn2file1 L3-L18

4.3 Problem Statement

The project aimed to convert basic C++ programming knowledge into working applications that accept user input and produce meaningful output. The three tasks addressed random-number guessing, menu-based arithmetic and task management.

4.4 Project Objectives

Strengthen practical understanding of C++ fundamentals.

Develop three working console-based applications.

Practice conditions, loops, functions, vectors, structures and switch-case statements.

Compile, test and demonstrate the programs.

Maintain the source code in a GitHub repository.

4.5 Scope of the Project

The scope covers three educational console applications. The projects demonstrate programming fundamentals and do not include graphical interfaces, databases, networking or persistent storage.

Number Guessing Game – random number generation, repeated guesses and feedback.

Simple Calculator – four basic arithmetic operations.

To-Do List – task addition, viewing, completion tracking and removal.

4.6 Technologies and Tools Used

Technology / Tool	Use
C++	Programming language for all three selected tasks.
Visual Studio Code	Editing, compiling and running the programs.
GCC / G++	C++ compiler.
MSYS2 UCRT64	Windows environment for the GCC/G++ toolchain.
GitHub	Storage and submission of source code.

4.7 System Architecture

The applications follow a simple console flow: user input → validation → processing/game logic → output. The calculator performs the selected arithmetic operation; the guessing game compares the guess with the generated number; the To-Do List manages tasks and their completion status.

This architecture is suitable for the educational, console-based nature of the projects.

4.8 Methodology

Stage	Activity
Requirement	Understand the task and required input/output.
Design	Plan the program flow and select suitable C++ constructs.
Implementation	Write the source code in Visual Studio Code.
Compilation	Compile using G++ and resolve configuration issues.
Testing	Run the programs with sample inputs and verify results.
Submission	Store source files in GitHub and record demonstrations.

4.8.1 Task 1 – Number Guessing Game

The program generates a random number from 1 to 100 and repeatedly asks the user for a guess. It gives high/low feedback until the correct number is found and counts the attempts. It uses `srand()`, `rand()`, a do-while loop and if-else conditions.

4.8.2 Task 2 – Simple Calculator

The calculator accepts two numbers and lets the user choose addition, subtraction, multiplication or division. A switch statement selects the operation and division by zero is checked before calculation.

4.8.3 Task 4 – To-Do- List

The To-Do List application allows users to add, view, complete and remove tasks. The implementation uses a vector of task records, loops, conditions and functions to manage task names and completion status.

4.9 Expected Outcomes

Outcome	Result
C++ Fundamentals	Improved practical use of variables, operators, conditions, loops and I/O.
Problem Solving	Converted task descriptions into step-by-step program logic.
Functions and Arrays	Applied modular logic and array-based board handling.
Testing	Compiled and tested programs using different user inputs.
Submission Skills	Organized source code and documentation in GitHub.

4.10 Offer Letter and Communication Proof

The offer letter supplied with the report confirms selection for the C++ Programming virtual internship with CodSoft and describes it as a one-month educational opportunity focused on learning, skill development and hands-on knowledge. The offer letter is dated 08 August 2026 and states the internship period as 10 August 2026 to 10 September 2026.



CodSoft

INTERNSHIP OFFER LETTER

Date : 08/08/2026

ID: BY26RY229027

Dear **Abhisar Sharma**,

We would like to congratulate you on being selected for the **“C++ Programming”** virtual internship position with **“CodSoft”**. We at CodSoft are excited that you will join our team. The duration of the internship will be of **1 Month**, starting from **10 August 2026 to 10 September 2026**. The internship is an educational opportunity for you hence the primary focus is on learning and developing new skills and gaining hands-on knowledge. We believe that you will perform all your tasks/projects. As an intern, we expect you to perform all assigned tasks to the best of your ability and follow any lawful and reasonable instructions provided to you.

We are confident that this virtual internship will be a valuable experience for you, we look forward to working with you and helping you achieve your career goals.

By accepting this offer, you commit to executing assigned tasks diligently and ensuring excellence in all aspects of your work.

Best of Luck!

Sincerely



Ruman Arshad
Program Manager



ISO Number - TSNUK79053



MSME Regd



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CodSoft Internship Offer Letter

GitHub

Repository:

https://github.com/abhi5ar/CODSOFT_TASKSNO

The three project demonstrations were also recorded and posted individually on LinkedIn as part of the internship submission process.

5. BIBLIOGRAPHY / REFERENCES

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Stroustrup, B. The C++ Programming Language. Addison-Wesley.